TFPT — Research Contracts for the Remaining Interfaces:

v_{geo}, G_{net}, F_{transfer}

(U_{wall}): the parabolic flavor wall-selection and (G_{metric}): the full quantum-gravity measure

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The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

You are reading companion R (Research Contracts).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\mathrm{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$	[E]/[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels $(U_{\mathrm{wall}}, G_{\mathrm{metric}}, F_{\mathrm{frontier}})$ are retained only for ledger continuity — the live residual is $v_{\mathrm{geo}} \oplus G_{\mathrm{net}} \oplus F_{\mathrm{transfer}}$.

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven   [C] conditional   [O] open / axiom   [X] kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\mathrm{car}} = 5$ *(two inputs, [O] axioms)*

Anchor $a = (1, 1, 2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

Physical bridges: v_{geo}, G_{net}, F_{transfer} *([C] bridges)*

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

What this document is about

After the compiler closure the entire residual is

$$\boxed{\mathrm{Rest} = (U_{\mathrm{wall}}) \oplus (G_{\mathrm{metric}}) \oplus (F_{\mathrm{frontier}})}$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. (F_{frontier}) (Koide, η_B, axion relic scale, m_p/m_e) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** (U_{wall}) *first* (finite, algebraic, falsifiable), then (G_{metric}) (deep analytic programme).

Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity). The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\boxed{\mathrm{Rest}_{\mathrm{operativ}} = v_{\mathrm{geo}} \oplus G_{\mathrm{net}} \oplus F_{\mathrm{transfer}}}$$

— $(U_{\text{wall}}) \to v_{\text{geo}}$: the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71), U_{point} collapses to the *one* dimensionful unit v_{geo} shared with $1/G$ (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where U_{point} still appears); $(G_{\text{metric}}) \to G_{\text{net}}$: holographically reduced to the single boundary-net statement *"the seam–Calderón inclusion has Jones index $4 = |\mu_4|$"* — holomorphy, $(E_8)_1$ and the unique 2D bulk then *follow* (v89/v92/`GATE.METRIC.05-07`); and the carrier-side QBL input now merges into the *same* statement: the tower carrier $\to SO(16)_1 \to (E_8)_1$ carries one and the same 16-fermion content, whose single quasi-free kernel (rank $= c$; Wick/Pfaffian exact) is the certified Calderón datum — closing G_{net} therefore also closes the carrier choice [E] ([`verification/v113_quasifree_kernel.py`]); $(F_{\text{frontier}}) \to F_{\text{transfer}}$: named QFT/cosmology *transfer* interfaces, never compiler powers.

Research Contract 1 — (U_{wall})

Goal. Prove

$$\nabla^{\star}_{F} = \mathrm{Sel}_{D_4,\,\mathrm{cusp},\,\det R=8,\,\mathrm{Spec}(Q_+)=\{1,2,3\}}\big(\mathcal{M}_{\mathrm{par}}(\mathbb{P}^1, \mu_4, E, \alpha)\big),$$

with $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ and $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$ at the four points of μ_4. The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one* D_4-symmetric, rank-3, four-point point that realises the selectors.

Lemma 1 (U0 — normalisation). *The gate data are equivalent to the $SU(3)$ character variety $\mathcal{X}_{\text{cusp}} = \{(M_1, \dots, M_4) \in C^4_{\text{cusp}} : M_1M_2M_3M_4 = \mathbf{1}\}/SU(3)$ with $C_{\text{cusp}} = \mathrm{Ad}_{SU(3)}\,\mathrm{diag}(1, \omega, \omega^2)$, $\omega = e^{2\pi i/3}$, and $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \dots, \gamma_4 \mid \prod \gamma_i = 1\rangle$. To show: $\mathcal{M}^{D_4}_{\text{par}} \simeq \mathcal{X}^{D_4}_{\text{cusp}}$ as polystable Mehta–Seshadri / tame Hodge realisation.* Machine: the group presentation and D_4-action are finite and codable (Sage/Wolfram).

Finite rigidity — now proven [E]. The finite half of U0 is closed exactly: μ_4 has cross-ratio 2 and Möbius stabiliser $\langle z \mapsto iz,\, z \mapsto 1/z\rangle$ of order $8 = |D_4|$ (a 4-cycle of the marks + a reflection, faithful); $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1}dz/(z^4 - 1)$, $k = 1, 2, 3$, are μ_4-eigenforms ($z \mapsto iz : \omega_k \mapsto i^k\omega_k$) carrying the three nontrivial characters χ_1, χ_2, χ_3 with weights $(1, 2, 3) = \mathrm{Spec}(Q_+)$. Hence a genus-zero boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$ with the $(1, 2, 3)$ grading (`QGEO.RIGID.01`, [E], [`verification/v168_qgeo_rigidity.py`]). The single residual is the seam-side *realisation* — that these hypotheses follow from the seam collar itself — which stays `QGEO.REALIZE.01` ([C]).

Seam Collar Realisation Theorem — the one central proof obligation [E]/[O] ([`verification/v176_seam_collar_realisation.py`])

Stated as one theorem and certified as a reduction — not a fabricated proof.

Theorem (target). *Given* (1) an oriented one-sided RP seam collar Σ; (2) a sheet-odd Calderón involution; (3) quasi-free one-particle seam data with gap $(2/3)^6$; (4) a faithful D_4 action on the boundary marks; (5) four parabolic marked points with μ_4-character decomposition; (6) an admissible c=8 boundary net — *then* the conformal boundary of Σ is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$, its H^1 character basis is canonically the generation basis with grading $(1, 2, 3)$, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$.

The proof decomposes into six steps, five already [E]:

- **1–2** *Geometry / uniformisation* [E] (v168): μ_4 has cross-ratio 2 and a faithful Möbius D_4 stabiliser of order 8; four genus-0 marks with faithful D_4 force the μ_4 square up to Möbius; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$.
- **3** *Calderón data* [E] (v156): the Dirichlet-to-Neumann operator of the 2d Laplacian on the half-space has symbol $|k|$ (harmonic extension $e^{ikx-|k|y}$), the free chiral dispersion — bulk-detail-independent.
- **4** *H^1 character = generation grading* [E] (v137/v168): $\omega_k = z^{k-1}dz/(z^4 - 1)$ carry μ_4 characters i^k with weights $(1, 2, 3) =$ the A_3 exponents $= \mathrm{Spec}(Q_+)$ — natural, not merely isomorphic.
- **5** *One-particle contraction* [E] (v162): a μ_4-equivariant operator is diagonal in the cusp basis

($\frac{1}{|\mu_4|}\sum_j U^j X U^{-j} = \operatorname{diag} X$ for $U = \operatorname{diag}(1, i, -1, -i)$); its spectrum is the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$, so the transfer eigenvalues $(1-\alpha)^{p_2}$ (p_2=6) are $\{1, (2/3)^6, (1/3)^6\}$, gap $(2/3)^6$ — forced, no free knob.

6 *AQFT closure* **[E]** (v154/v175): $(D_5)_1 \otimes (A_3)_1$ with the isotropic μ_4 glue has index $4 = |\mu_4|$, $c = 5+3 = 8$, μ-index $16/16 = 1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ ($248 = 120+128$); full-cone RP holds for all m via the CAR second-quantisation functor.

The proof tree — and the *two* genuine open doors [O] ([`verification/v177_seam_marking_kernel.py`]). Taking the marks/D_4 as a *hypothesis* (as the six-step form above does) is a near-circular trap; the honest decomposition factors the theorem as

$$\textsc{Realize} = \textsc{Marks} \cdot \textsc{Uniform} \cdot \textsc{Cohom} \cdot \textsc{Module} \cdot \textsc{Kernel} \cdot \textsc{Net},$$

of which *four nodes are now closed constructively*: Uniform **[E]** (an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 — so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P}^1, \mu_4)$, strengthening v168's cross-ratio to the full uniformisation); Cohom **[E]** ($\rho^*\omega_k = i^k\omega_k$, grading $(1,2,3)$); Module **[E]** (a *unique* μ_4-equivariant iso — multiplicity-free + residue normalisation, reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); and Net **[E]** (v154/v175). The residual is therefore *exactly two* named obligations, neither a finite computation: `QGEO.MARKS.01` — the *raw* RP seam collar canonically produces the genus-0 four-parabolic-marked D_4 boundary (the four marks from the anchor atom $e_1(a)$=4=$|\mu_4|$, collisions excluded, $\tau\rho\tau$=ρ^{-1}) — and `QGEO.KERNEL.01` — the *raw* seam Calderón operator equals the μ_4-equivariant free c=8 gapped one-particle contraction *as an operator* ($C_\Sigma = U^{-1} C_{\mu_4} U$), not merely the same spectral list. **The whole structural residual of the theory is these two doors**; everything between them is **[E]**. They need a human constructive-geometry/AQFT argument (or a proof assistant), not more arithmetic.

How far each door opens by computation ([`verification/v178_marks_kernel_attempt.py`]). Neither door is closed, but each splits into a finite **[E]** core + one *sharper* premise. *For* Marks*: given* a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1, i, -1, -i\} = \mu_4$ (not the fixed points), distinct iff $b_1 = 3 = N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving faithful D_4 — so Marks reduces to the *single* topological premise "the seam double is genus-0 with an order-4 clock whose marks form one orbit". *For* Kernel*:* on the 3-dim multiplicity-free H^1 the μ_4 characters $(1,2,3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and is *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ and the unique residue-normalised basis are *equal as operators*, not merely isospectral; the "spectrum vs operator" distinction *vanishes* on the finite block. So Kernel reduces to the *single* premise "the full DtN on L^2 reduces to this 3-dim gapped transfer block + the fixed rank-8 polarisation" (principal symbol $|k|$, Lee–Uhlmann v156; no drift v158). The two doors are thereby *milder*, but still open.

And the two doors are in fact ONE ([`verification/v179_conformal_realisation.py`]). Marks's residual is not even three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\, 2\pi\, \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$, and $\chi = 2 - 2g \Rightarrow g = 0$, so "genus-0" is the *same* $\chi = 2$ that supplies the "8" in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — it is the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* — the parabolic marks are not the two elliptic fixed points $\{0, \infty\}$, and a free order-4 orbit has exactly $4 = e_1(a)$ points. So Marks reduces to "the carrier clock acts *conformally* on the genus-0 double", and Kernel then *follows* from that same geometry (Lee–Uhlmann $|k|$ + the $c = 8$ rigidity v158 + the cusped-surface spectral split, residual $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$). **Hence the entire structural residual of the theory is a single geometric premise** `QGEO.CONF.01`: *the raw RP seam normal double is conformally* $(\mathbb{P}^1, \mu_4)$ *with the carrier clock as the order-4 Möbius automorphism* $z \mapsto iz$. Given it, everything — genus-0, the four D_4 marks, the $(1,2,3)$ grading, the DtN block, the rank-8 polarisation and the $(E_8)_1$ net — follows. This single premise is a constructive-geometry statement (the conformal class of the raw seam *is* the cusped sphere), *not* a finite computation, and is left honestly **[O]**.

Lemma 2 (U1 — wall landscape, *corrected*)**.** *Parabolic degree zero forces the line weight-sums onto the wall* $(w_1, w_2, w_3) = (2, 1, 1)$. *An explicit balanced representative (weights in* $\frac{1}{3}$*) is*

$$W_{\text{wall}} = \tfrac{1}{3}\begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

each column a permutation of $(0,1,2)$*, row sums* $(2,1,1)$*.* ***Honest result:*** *the* $6^4{=}1296$ *column-permutation matrices give* 144 *walls; under the* full *group* $D_4 \times S_3(\text{lines}) \times S_3(\text{values})$ *they form* five *orbits,* not one*. So* W_{wall} *is* not *singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set;* W_{wall} *is one of the five. Machine: fully certified* $C_U^{(1)}$ *(*[verification/v29_research_contract_certs.py]*).*

The splitting type is now algebraic (RH route, [verification/v32_rh_splitting.py]***).*** *Passing from the monodromy to the Fuchsian* residues A_k *(eigenvalues = weights* $\{0,\frac{1}{3},\frac{2}{3}\}$*,* $A_k = U^{k-1}A_0U^{-(k-1)}$*,* $U = \mathrm{diag}(1,i,-i)$*) the twisted* $\mathbb{Z}_4$*-average collapses* exactly*:* $\sum_k U^kA_0U^{-k} = 4\,\mathrm{diag}(A_0)$*, so the exponents at infinity are* $4\,\mathrm{diag}(A_0)$*. A flat bundle with regular* ∞ *needs integer exponents summing to* $4{=}-\deg E$*; with the cusp weights the only options are perms of* $(2,1,1)$ *and* $(2,2,0)$*, i.e.*

$$\boxed{\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2 \iff \mathrm{diag}(A_0) = (\tfrac{1}{2},\tfrac{1}{4},\tfrac{1}{4})}$$

(the sibling $(2,2,0) \to \mathcal{O}(0)\oplus\mathcal{O}(-2)^2$*). Such a Hermitian* A_0 *(spectrum* $\{0,\frac{1}{3},\frac{2}{3}\}$*, that diagonal) exists by Schur–Horn. So the* splitting *is an algebraic condition on* $\mathrm{diag}(A_0)$*.* Honest wall: *the global* $\prod M_k = \mathbf{1}$ *is the path-ordered monodromy (*not $\exp(2\pi i\sum A_k)$*), constraining* off-diag(A_0) *via the genuine RH solve; and R-extraction still needs the* $C_6 \leftrightarrow \mathbb{P}^1\setminus\mu_4$ *bridge. The RH route fixes the* splitting *algebraically but not* c_u/c_d*.*

Lemma 3 (U2 — the kill switch, *run and hardened* [E])**.** *The selected polystable wall representative must* not *collapse to the diagonal abelian* $U(1)^3$ *point if it is to give non-trivial* $SU(3)_F$ *transport mixing:* ***(A)*** $\nabla_F^\star$ *polystable but* not *simultaneously diagonal (the desired breakthrough), or* ***(B)*** *diagonal collapse, in which case* c_u/c_d *stays honestly [C].* ***Result: the cheap test is run robustly and lands on A.*** *The kill-switch fires iff the* D_4*-symmetric cusp representation is* reducible *(commutant non-scalar). Across* 400 *random cusp-class generators of the* $\mathbb{Z}_4$ *family* $M_k = U^{k-1}MU^{-(k-1)}$ *the commutant is scalar at* every *point (*irreducible $400/400$*), the reducible locus is hit with probability* 0 *(measure-zero, codimension* ≥ 1*), and the mixing strength stays bounded away from zero (*min *off-diag* $= 0.35 > 0.05$*). So case A is the* generic *behaviour, not an accident of the one explicit point of* [verification/v33_explicit_flat_bundle.py]*:* (U_{wall}) survives *its own cheapest falsification test.* [verification/v38_uwall_killswitch.py]

Lemma 4 (U3 — D_4 fixed locus as an explicit variety, *positive-dimensional*)**.** *With* $M_k = U^{k-1}AU^{-(k-1)}$ *(*$U = \mathrm{diag}(1,i,-i)$*, the* $\mathbb{Z}_4$ *rotation) and the reflection* $M_1 = VM_1^{-1}V^{-1}$*,* $M_3 = VM_3^{-1}V^{-1}$*,* $M_2 = VM_4^{-1}V^{-1}$ *(with* $V = -[[1,0,0],[0,0,1],[0,1,0]]$*,* $V^2{=}1$*,* $VUV^{-1}{=}U^{-1}$*), the relations become polynomials in the entries of* $A \in C_{\text{cusp}}$*;* $\mathcal{X}_{\text{cusp}}^{D_4}$ *is cut out as* $\bigcup_k C_k$*, parametrised by trace coordinates* $x = \mathrm{tr}(M_1M_2)$ *(note* $\mathrm{tr}\,A = 1{+}\omega{+}\omega^2 = 0$*).* ***Result (***$C_U^{(2)}$***,*** [verification/v30_d4_character_variety.py]***):*** *adding the reflection to the* $\mathbb{Z}_4$ *locus of* v19 *does* not *isolate the point — the* full D_4*-fixed product locus is still* positive-dimensional *(*$|\mathrm{tr}(M_1M_2)|$ *varies continuously over* $\approx [0,3]$*). So even the full* D_4 *symmetry cannot select* $\nabla_F^\star$*.* Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).

Lemma 5 (U4 — selectors as equations)**.** *The readouts* $\rho \mapsto \det R(\rho)$*,* $\rho \mapsto \mathrm{Spec}(Q_+(\rho))$*,* $\rho \mapsto \Lambda_{f,j}(\rho)$ *are well-defined (cyclotomic/algebraic) on* $\mathcal{X}_{\text{cusp}}^{D_4}$*.* ***Acceptance criterion:***

$$\#\{\rho \in \mathcal{X}_{\text{cusp}}^{D_4}:\ \det R = 8,\ \mathrm{Spec}(Q_+) = \{1,2,3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not *closed.* ***The bottleneck (made precise by U3):*** *since the* D_4*-fixed variety is positive-dimensional, the* selector *must do all the cutting — and evaluating* $\det R(\rho)$*,* $\mathrm{Spec}(Q_+(\rho))$ *as functions on* $\mathcal{X}_{\text{cusp}}^{D_4}$ *requires the* parabolic-degree $\leftrightarrow$ residue dictionary $R(\rho)$*, i.e. the H2 equivalence. That dictionary was the single remaining analytic input of the* (U_{wall}) *gate.* Status update: since v118–v122 the dictionary itself is exact — *hexagon = sign-twisted family spectrum, lepton coefficients*

= *resolvent determinants of the $W(A_3)$ monodromy, addresses pinned, margins theorems (boxes below); what remains is the strictly smaller $R4'$, the* establishment of the selectors *($n = (5, -9, 6)$ + the diamond determinants; since [`verification/v134_dual_anchor.py`]: the pair (d, n)).*

***What** $R(\rho)$ **is** (*[`verification/v31_R_dictionary.py`]*).* *Two facts pin its nature.* (i) Case A is alive: *every D_4-fixed cusp monodromy found is* irreducible*, so the wall point carries non-trivial $SU(3)_F$ mixing — the U2 kill switch does* not *fire to the diagonal case B.* (ii) $R(\rho)$ is not algebraic: *R is integer-valued, but the algebraic invariants of ρ (e.g.* $\mathrm{tr}(M_1M_2)$*) vary* continuously *on the locus; an integer readout cannot be a continuous algebraic function. Hence $R(\rho)$ is the discrete invariant of the* non-abelian Hodge / Mehta–Seshadri *map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle E_ρ): integer output, but its* computation *is the transcendental harmonic-metric (Hitchin) solve* on the generic Higgs locus. ***Refinement** (**$U4''$**, [`verification/v40_harmonic_metric.py`]):* *at the* polystable *point TFPT actually selects this pessimism does not apply — see the box below.*

Progress: the *selector side* of U4 is closed on the explicit point ([E]). While $R(\rho)$ as a function on the whole locus needs the Hitchin solve, the two U4 *selectors* are *algebraically* accessible on the explicit flat bundle of [`verification/v33_explicit_flat_bundle.py`] — no PDE. Computed exactly from A_0: the exponents at infinity $\mathrm{eig}(\sum_k A_k) = \{2, 1, 1\}$ give the splitting type $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$, i.e. the parabolic *anchor* $a = (1, 1, 2)$; the parabolic degree is 0 ($\deg E = -4 = -|\mu_4|$ plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\boxed{\begin{array}{c}\det R = 8 = n\cdot a \quad \text{(anchor pairing on the splitting type)},\\ \mathrm{Spec}(Q_+) = \{1, 2, 3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\})\end{array}}$$

So $\det R = 8$ is the anchor pairing of the splitting type and $\mathrm{Spec}(Q_+)$ is the affine image (d=3 the weight denominator) of the cusp weights — *both selectors are read off the explicit bundle's algebraic data.* **Residual (sharpened below by Readout Rigidity):** the quark *ratio* c_u/c_d turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation (U_{point}), which is an anchor, not a missing PDE. [`verification/v39_uwall_selectors.py`]

Breakthrough: the harmonic metric is *finite linear algebra*, not a PDE ([E]). The U4 "transcendental Hitchin solve" is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field $\Phi = 0$. So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [`verification/v33_explicit_flat_bundle.py`] this is realised: the raw monodromies M_k are non-unitary ($\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$), but there is a *unique* common invariant Hermitian form H ($\dim\ker\{M_k^\dagger H M_k = H\} = 1$), it is *positive-definite*, and the unitarised holonomy $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$ is unitary ($\sim 10^{-8}$), lies in the cusp class $\{1, \omega, \omega^2\}$ with $\prod \widetilde{M}_k = \mathbf{1}$, and has *clean* matrix elements $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$. **So $C_U^{(3)}$ reduces from a transcendental PDE to: this finite unitarisation (done) + the combinatorial word-length R (closed: H1 + $\det R = 8$).** *Residual:* the feared PDE is gone; the quark *ratio* c_u/c_d is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [`verification/v40_harmonic_metric.py`]

The final leg-assignment test (honest result) ([E]). Does the harmonic-frame holonomy $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ feed the $\delta = \frac{1}{2}$ resolvent to give the lepton amplitudes $\Lambda = (0.475, 1.107, 0.917)$? **(i) Ruled out as a literal identity:** $\Lambda_\mu = 1.107 > 1$, while unitary holonomy entries have modulus ≤ 1, so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly $|\mathrm{diag}\,\widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, and that $\frac{1}{2}$ *equals* the distinguished lepton transport value $\delta = \frac{1}{2}$ that v20 used by hand — so the geometry plausibly *supplies* δ (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g. $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the $\delta = \frac{1}{2}$ resolvent (closed combinatorially via the word-length R); the holonomy supplies δ and the leg selection, not the amplitude magnitude. c_u/c_d is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ($\delta = \frac{1}{2}$). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [`verification/v41_leg_assignment.py`]

The torsion normal form and the $\delta = \frac{1}{2}$ theorem ([E]). The v41 "genericity is future work" item is now executed — and yields a theorem. *(i) Flatness = μ_4 torsion [E]:* for *any* M (symbolic), $\prod_k U^k M U^{-k} = (MU)^4$ — the $\mathbb{Z}_4$-family flatness constraint *is* the torsion statement "$T = MU$ is a fourth root of unity": the μ_4 atom appears as the *order* of the twisted cusp generator. *(ii) The $\delta = \frac{1}{2}$ theorem [E] (both directions):* on the *involutive* branch ($T^2 = \mathbf{1}$) T is a Hermitian reflection $2vv^\dagger - \mathbf{1}$, and the cusp trace condition $2v^\dagger U^{-1} v = 1$ splits *exactly* into $|v_1|^2 = \frac{1}{2}$ (real part) and $|v_2| = |v_3|$ (imaginary part), forcing

$$\boxed{\operatorname{diag} M = (0, \tfrac{i}{2}, -\tfrac{i}{2}), \qquad \operatorname{spec} M = \{1, \omega, \omega^2\} \text{ automatic}}$$

($\chi_M = \lambda^3 - 1$ from unitary + trace 0 + det 1). So $\delta = |M_{22}| = \frac{1}{2}$ holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a *μ_4-torsion identity*, not an observation. *(iii) Reflection lemma [E]:* the D_4 reflection forces $\operatorname{diag} M = (r, z, \bar{z})$ with r real and $\operatorname{Re} z = -r/2$ — the whole D_4 diagonal freedom is *one* real parameter, and $\delta = \frac{1}{2} \iff r = 0$. *(iv) Branch census [E] (seeded):* the full D_4-fixed flat cusp locus realises only $\operatorname{tr}(MU) \in \{1, -1\}$; the involutive branch has the diagonal *constant* $(0, \frac{1}{2}, \frac{1}{2})$ to machine precision, while on the $\{1, i, -i\}$ branch d_1 varies over $[0, 1]$ — and the explicit v33/v40 point sits exactly in its $d_1 = 0$ slice, the *same* δ value. *Residue [C] (recorded):* whether the anchor splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (Mehta–Seshadri side) forces the $d_1 = 0$ slice — or selects the involutive branch — is the remaining question; $\delta = \frac{1}{2}$ itself is no longer the open item. [`verification/v114_torsion_delta.py`]

The anchor pins the residue: an exact normal form for the (U_{wall}) bundle ([E]). The v114 branch question is answered — and the v33 *numerical* Riemann–Hilbert point acquires an *exact closed form*. *(i) μ_4-average lemma [E]:* $\sum_k U^k X U^{-k} = 4 \operatorname{diag} X$ for any X — μ_4 conjugation-averaging *is* the diagonal projection, so the exponents at infinity are $4 \operatorname{diag} A_0$ and the anchor splitting holds *iff* $\operatorname{diag} A_0 = (2, 1, 1)/4 = a/|\mu_4|$: the residue diagonal **is** the anchor over μ_4 (v33's diagonal was forced, not chosen). *(ii) The $(8, 0, 5)/144$ lemma [E]:* the anchor diagonal and the cusp spectrum $\{0, 1, 2\}/N_{\text{fam}}$ pin $\sum |a_{ij}|^2 = \frac{13}{144}$, and with $a_{13} = 0$ the determinant condition solves *uniquely* to

$$\boxed{(|a_{12}|^2,\ |a_{13}|^2,\ |a_{23}|^2) = \frac{(8,\ 0,\ 5)}{144}} \qquad \begin{array}{l} 8 = \operatorname{rank} E_8,\ 5 = g_{\text{car}}, \\ 144 = (|\mu_4| N_{\text{fam}})^2,\ 8+5 = 13 = \Delta_Q \end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same* Δ_Q that denominates $c_u/c_d = 55/117 = 5 \cdot 11/(9 \cdot 13)$. *(iii) Exact normal form [E]:* $A_0^\star = \left[\begin{smallmatrix} 1/2 & \sqrt{2}/6 & 0 \\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12 \\ 0 & \sqrt{5}/12 & 1/4 \end{smallmatrix}\right]$ has $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ *exactly*. *(iv) $A_0^\star$ is the RH solution [E]:* its big-circle monodromy is trivial to 10^{-10}, $\prod M_k = \mathbf{1}$, and the unitarised holonomy has $|\operatorname{diag} \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, $\operatorname{tr}(\widetilde{M}_0 U) = 1$ — the v33/v40 point is gauge-equivalent to $A_0^\star$ (the a_{12}, a_{23} phases are diagonal gauge). *(v) The anchor forces the $d_1 = 0$ slice [E]:* a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with $|\operatorname{diag} \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ — combined with v114, the whole harmonic diagonal is **anchor + torsion forced**; $\delta = \frac{1}{2}$ needs no selection. *Residue [C]:* $a_{13} = 0$ and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [E] via Gröbner on the RH constraints is the remaining U_{point} step. [`verification/v115_anchor_residue.py`]

The resonance theorem: $M_\infty = \mathbf{1} \iff a_{13} = 0$ — one gauge orbit [E] ([`verification/v116_resonance_uniqueness.py`])

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as $dY/dw = -(1/w)(B_0 + B_1 w + \cdots)Y$ with $B_m = \sum_k i^{km} U^k A_0 U^{-k}$, the *twisted* μ_4 averages collapse [E]:

$$B_0 = 4\,\mathrm{diag}\,A_0, \qquad \boxed{B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}}$$

(only the eigenvalue-ratio $-i$ cells survive). The anchor exponents $\mathrm{diag}(2,1,1)$ have resonance gap 1; the level-1 formal-gauge equation $(1 - \mathrm{ad}\,R_0)H_1 = R_1$ is singular exactly on the cells $\{(2,1),(3,1)\}$, where B_1 has entries $(0,\,4\overline{a_{13}})$; all higher levels are non-resonant and the formal monodromy is $e^{-2\pi i\,\mathrm{diag}(2,1,1)} = \mathbf{1}$. Hence [E]:

$$\boxed{M_\infty = \mathbf{1} \iff a_{13} = 0}$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation. Uniqueness corollary [E]:* $a_{13} = 0$ + the $(8,0,5)/144$ moduli + phases = diagonal gauge ⇒ the flat anchor locus is **exactly one gauge orbit**, the exact $A_0^\star$: within the μ_4-equivariant anchor class the (U_{wall}) bundle datum is *algebraically unique. Sharpness [E]:* $\|M_\infty - \mathbf{1}\| \sim 8.5\,|a_{13}|$ (0.17 at $a_{13} = 0.02$) vs 10^{-10} at $A_0^\star$. *Honest scope [C]:* the equivariant ansatz is the established D_4 selector input (v19/v30); the *residue* side is now [E], while the holonomy values (the harmonic diagonal $(0,\frac{1}{2},\frac{1}{2})$) remain [E] — transcendental monodromy of an exact system.

The monodromy is the Weyl group of A_3 — the holonomy is exact [E] ([`verification/v117_monodromy_weyl_a3.py`])

The last [E] item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact $A_0^\star$ system is itself *exact*, with entries in $\frac{1}{2}\mathbb{Z}[i]$:

$$\boxed{\widetilde{M}_0 = \begin{pmatrix} 0 & -\frac{1+i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2} \\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2} \end{pmatrix}}$$

unitary, $\det = 1$, $\mathrm{tr} = 0$, $\chi = \lambda^3 - 1$ (the cusp class *exactly*), $\widetilde{M}_0^3 = \mathbf{1}$, and $\mathrm{diag}\,\widetilde{M}_0 = (0, -\frac{i}{2}, +\frac{i}{2})$ — so $d_1 = 0$ and $\delta = \frac{1}{2}$ are now *matrix entries*, not observations; $(\widetilde{M}_0 U)^4 = \mathbf{1}$ with $\mathrm{tr}(\widetilde{M}_0 U) = 1$ (the $\{1, i, -i\}$ branch, $d_1 = 0$ slice — exactly where v114/v115 located the physical point). *The group* [E]: exact enumeration of $\langle U, \widetilde{M}_0\rangle$ gives order 24 with order statistics $(1,9,8,6)$ and character values $(3,-1,0,1)$ — uniquely S_4 in its standard ⊗ sign representation, i.e. the (twisted) reflection representation of the **Weyl group of the family lattice** A_3:

$$\boxed{\mathrm{monodromy}(U_{\mathrm{wall}}) = W(A_3) = S_4, \qquad |W(A_3)| = 4! = 24}$$

— the flavor wall realises the A_3 factor of the glue $D_5 \oplus A_3 + \mu_4$ as its Weyl group (U = a 4-cycle, the μ_4 deck; $\widetilde{M}_0$ = a 3-cycle, the family rotation). *Identification [E]:* the ODE monodromy of $A_0^\star$ equals the exact $\widetilde{M}_0$ to 3.5×10^{-10}, and the invariant form H is U-invariant to 10^{-11}. *Status:* holonomy values [E]; only the analytic identification stays [E]. The δ thread (v20 → v40/v41 → v114 → v115 → v117) is closed end to end.

The hexagon is the sign-twisted family spectrum — first exact H2 piece [E] ([`verification/v118_hexagon_family_dictionary.py`])

The missing "$C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary" (H2) acquires its first exact entry. *Sign-twist lemma [E]:* $-\widetilde{M}_0$ has order 6 and

$$\boxed{\mathrm{spec}(\widetilde{M}_0) \cup \mathrm{spec}(-\widetilde{M}_0) = \mu_6}$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ($\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ realised as $(-1) \times \langle \widetilde{M}_0 \rangle$); the v20 denominators $\frac{5}{4} - \cos(r\pi/3)$ are exactly the eigen-denominators $|1 - \zeta_6^r/2|^2$ of the two family resolvents $(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0)^{-1}$. *Cyclotomic determinant [E]:* $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 = (1-t)\Phi_3(t)$, so $\det(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$ — and the lepton coefficients become resolvent determinants: $c_e = |\mu_4|/(2\det_-) = \frac{16}{7}$, $c_\mu = N_{\rm fam}/(2\det_+) = \frac{4}{3}$, $c_\tau = |\mu_4|\det_-/N_{\rm fam} = \frac{7}{6}$, product rule $= |\mu_4|/\det_+ = \frac{32}{9}$. The lepton 7 and 9 *are* the family-resolvent determinants; $\delta = \frac{1}{2} = |(\widetilde{M}_0)_{22}|$ is supplied by the matrix itself. *Sheet-extended group [E] (audit):* $\langle U, \widetilde{M}_0, -\mathbf{1}\rangle$ has order $48 = \Omega_{\rm adm} = N_{\rm fam}\dim S^+$ — the seed's quartic coefficient. *Residue [C]:* the dictionary's *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains v20's established input / open — nothing re-fished.

The address table: the lepton words are the compiler atoms [E]/[C] ([verification/v120_address_table.py])

The address table acquires its structure — on word-length integers *frozen* in v18/v20, long before today's readings (the red-team firewall). *(i) The lepton words are the atoms [E]:*

$$\boxed{(L_e, L_\mu, L_\tau) = (8,5,3) = (\mathrm{rank}\,E_8,\ g_{\rm car},\ N_{\rm fam}) = (p_0{+}e_2,\ e_2,\ p_0)}$$

in mass order (longest word = lightest lepton). *(ii) Addresses = division by the hexagon [E]:* $(r,w) = L$ divmod p_2 with $p_2 = 6 = |R^+(A_3)|$ itself an atom: $e \to (2,1)$, $\mu \to (5,0)$, $\tau \to (3,0)$; $\sum r = 10 = p_3 = \mathcal{A}_\Lambda$, $\sum L = 16 = \dim S^+$. *(iii) Sheet parity [E]:* by the v118 dictionary, e sits on the *untwisted* sheet ($\omega \in \mathrm{spec}\,\widetilde{M}_0$), μ, τ on the *twisted* sheet $(-\omega, -1)$; the anchor lepton τ occupies the unique *real* twisted eigenvalue -1 — exactly the site where v20's product rule replaces the resolvent. *(iv) Quark sum rules [E] (frozen v18 words* $7,7,5,3,2,0$*):* up-type sum $= 10 = p_3$; down-type sum $= 14 = p_1{+}p_3$; **all quarks** $= 24 = |W(A_3)|$ (the monodromy group order, v117); **all nine charged fermions** $= 40 = p_1p_3 = a^TRa$ (v119); quark $\sum r = 2p_2$, $\sum w = |\mathbb{Z}_2|$; the top sits at the vacuum site $(0,0)$ — zero word, no suppression, the established ladder anchor. *Residue [C]:* five independent closures $(16, 10, 14, 24, 40)$ constrain the address map; the per-fermion *assignment* remains the v18/v20 input — deriving it is the remaining H2 content.

The address table is pinned: R is the unique hexagon matrix with atom margins [E] ([verification/v121_address_pinning.py])

The per-fermion assignment closes at the *information* level. The established identity $L = R{+}2U$ (v95) says the word-length table *is* L[sector][generation] — rows (up; down; lepton) $= ((7,3,0);(7,5,2);(8,5,3))$, the v18/v20 words verbatim — so the addresses are the entries of the *residue operator* R plus one hexagon turn for the first generation, and R's first column is the **anchor** $a = (1,1,2)$. The margins are atoms [E]: row sums $(4,8,10) = (e_1, \mathrm{rank}\,E_8, p_3)$, column sums $(4,13,5) = (e_1, \Delta_Q, g_{\rm car})$. And the census closes it [E]:

$$\boxed{\begin{array}{c}\text{among all } 3\times 3 \text{ matrices with entries in the hexagon } \{0..5\},\\ \text{exactly 17 have the atom margins — and exactly ONE has } \det = 8\end{array}}$$

— and it is R (also the unique one with SNF $= (1,1,8)$; all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ($m \sim \lambda_Y^L$: longer word = lighter fermion). *Residue [C]:* the open H2 item shrinks from "derive 9 per-fermion integers" to "derive the six atom margins + det = rank". *Firewall:* R, L and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

The margins are theorems ([verification/v122_margin_theorem.py]). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed — *(S1)* the D_4 annihilator $n = (5,-9,6)$ with $n^TR = (8,0,0)$ (it kills the unchanged (c_2, c_3) plane; $n\cdot a = 8$), *(S2)* $\det R = 8$, *(S3)* $\det K = 4$ with $K = R + Q\Sigma$ — pin R *uniquely* among hexagon matrices [E]: the $(S1)$ census has $4\times4\times4 = 64$ candidates, $(S2)$ leaves 12, $(S3)$ leaves **exactly one**, and it is R. Hence the atom margins, the anchor column $Re_1 = a$ and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [E]:* on the $(S1)+(S2)$ locus $\det(M+2U) = 20 = \det L$ holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler's familiar overdetermination. *Residue [C]:* H2 reduces to the establishment of the selectors themselves (n and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class.*

The dual-normal selector: (d,n) pins R columnwise

([`verification/v136_dual_normal_selector.py`]). The selector list shrinks once more. With the dual anchor $d = a^\top R^{-1}$ ([`verification/v134_dual_anchor.py`]) and the torsion normal n, each *column* of R obeys $d \cdot c_j = a_j$ and $n \cdot c_j = 8\delta_{1j}$; the common kernel of (d, n) is the primitive vector $(6, 8, 7)$ (audit: p_2, rank E_8, scalaron-7), so within the address box $\{0..5\}^3$ each column is the *unique* lattice point (census: 1 of 216 each) — $(d, n) \Rightarrow R$, and the 6^9 census above collapses to three column problems [E]. *Honesty:* K is *not* pinned directly ($d^\top K = (1, \frac{1}{2}, 1)$); $\det K = 4$ comes via the μ_4 wall ([`verification/v135_det_surface.py`]). *The mechanism is bounded:* the zero mode of $A_0^\star$ carries the exact spectral normal $\sigma = (2, -9, 5) = (|\mathbb{Z}_2|, -N_{\rm fam}^2, g_{\rm car})$ (signed squares; $\|v\|^2 = \frac{16}{9}$ with $16 = \dim S^+$, $\operatorname{adj} A_0^\star = \frac{2}{9} P_v$ with $\frac{2}{9}$ = the SdS entropy curvature), $\sigma \cdot \mathbf{1} = -|\mathbb{Z}_2|$, $\sigma \cdot a = N_{\rm fam}$, $n \cdot \sigma = 121 = \|\operatorname{Pl}(K)\|_1^2$; and the exact $W(A_3)$ orbit control shows *no* group image of σ is proportional to n — the naive cofactor/orbit lift fails, the step $\sigma \to n$ is a genuine mechanism beyond the monodromy group. *R4′ restated [C]:* derive the *pair* (d, n); then R is forced columnwise.

The selector triangle ([`verification/v139_selector_triangle.py`]). The pair itself now collapses further. First, d is *pure anchor data*: $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ exactly — the closed form never mentions R, so the first selector is *derived*, not residual. Second, the frame $(\mathbf{1}, a, \sigma)$ is a basis of $\mathbb{Z}^3$ with $\det(\mathbf{1}|a|\sigma) = 11 = \|\operatorname{Pl}(K)\|_1$ (the quark constant as the frame volume), and n is the *unique* integer covector with the three atom pairings

$$\boxed{n \cdot \mathbf{1} = 2 = |\mathbb{Z}_2|, \qquad n \cdot a = 8 = \operatorname{rank} E_8, \qquad n \cdot \sigma = 121 = \|\operatorname{Pl}(K)\|_1^2}$$

— the cofactor definition is replaced by frame data plus three atoms. On the $(\mathbf{1}, a)$-constrained line the σ-pairing is $11(11{+}t)$: divisible by 11 for *every* t, a perfect square exactly at the true n ($t{=}0$). The review's predicted lift is exact as an audit identity: $n = \operatorname{reverse}(\sigma) + |\mu_4| e_3$ (generation flip + μ_4 lift at the third slot).

Frame integrality: three pairings → two ([`verification/v142_frame_integrality.py`]). The third pairing is *not* independent. The pairing triples $(m \cdot \mathbf{1}, m \cdot a, m \cdot \sigma)$ of *integer* covectors form an index-11 sublattice of $\mathbb{Z}^3$, cut out by the single congruence

$$\boxed{x - 3y + z \equiv 0 \pmod{11}} \qquad (\text{index } 11 = \|\operatorname{Pl}(K)\|_1 = \det(\mathbf{1}|a|\sigma)),$$

so with the two line pairings at their atoms $(x, y) = (|\mathbb{Z}_2|, \operatorname{rank} E_8) = (2, 8)$, integrality *forces* $z \equiv 0 \pmod{11}$ — the $11(11{+}t)$ line *is* the integer line. Primitivity forces t *even* (odd t makes every entry of $n(t)$ even), and $t{=}0$ is the unique even t with a square σ-pairing for $|t| < 88$ (the global pick keeps the square selector — recorded honestly). Both line pairings are Cramer-reducible to inventoried identities ($n \cdot a = \det R$ by Laplace along the anchor column; $n \cdot \mathbf{1} = 8(R^{-1}\mathbf{1})_1$ with $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$, v134) — restructuring, not establishment.

The pairing values are atom identities ([`verification/v145_pairing_atoms.py`]). Even the two line-pairing *values* carry no independent information. Reading the v139 lift structurally — $n = w_0(\sigma) + |\mu_4| e_3$ with w_0 the A_3 *exponent duality* $m \leftrightarrow h{-}m$ (an established Lie involution; lift slot = the top-exponent line, lift size = the glue order) — all three pairings become atom arithmetic:

$$n{\cdot}\mathbf{1} = |\mu_4| {-} |\mathbb{Z}_2| = |\mathbb{Z}_2|, \qquad n{\cdot}a = |\mu_4|\, a_3 = 8 = \operatorname{rank} E_8 \ \text{ via } \boxed{|\mu_4| + g_{\rm car} = N_{\rm fam}^2 \quad (4 + 5 = 9,\ \sigma \perp w_0(a))},$$

and $n \cdot \sigma = \|\sigma\|^2 - N_{\rm fam}^2 + |\mu_4| g_{\rm car} = 110 - 9 + 20 = 121$ with $\|\sigma\|^2 = |\mathbb{Z}_2|^2 + N_{\rm fam}^4 + g_{\rm car}^2 = 110 = |\mathbb{Z}_2| g_{\rm car} \|\operatorname{Pl}(K)\|_1$. *R4′ residue [C] (final form):* derive the *one* map $n = w_0(\sigma) + |\mu_4| e_3$ — no independent number remains; everything else in the chain anchor $\to (d, n) \to R$ is exact.

Both dual normals are anchor data ([`verification/v149_cusp_normal.py`]). The remaining opacity dissolves in the canonical frame: the pairings of n with the *cusp eigenbasis* (the geometric basis of v98) are exactly

$$\boxed{n \cdot e_1 = 6 = p_2(a), \qquad n \cdot e_2 = 3 = p_0(a), \qquad n \cdot e_3 = 5 = e_2(a)}$$

— one anchor invariant per Q_+ degree, and the cusp matrix is unimodular, so these three pairings determine n uniquely. With $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ the dual normal *pair* (v134) is anchor data through and through — one normal per frame; the cusp data $(6, 3, 5)$, the frame pairings $(2, 8, 121)$ and the w_0-lift are three presentations of one covector, all atomic (equivalences exact). *Audit:* $\sigma \to (5, 1, 2)$; alternate reading $(6, 3, 5) = (|R^+(A_3)|, N_{\rm fam}, g_{\rm car})$; sum $14 = \dim G_2$. *R4′ residue [C] (cusp form):* derive the *assignment* — why the Q_+ degrees $(1, 2, 3)$ carry (p_2, p_0, e_2) in that order; one discrete assignment, zero continuous freedom.

Exterior Leg Lemma — the quark u/d leg is a $\Lambda^2 F$ area, not a scalar [E]/[C]

The v41 negative is a *category error*, not a dead end. u and d share the hexagon-radial data $(r, w) = (1, 1)$ ($L = 7$ for both), and the C_6 resolvent $c(r, w) = |\mu_4|^w/(\frac{5}{4} - \cos\frac{r\pi}{3})$ reads *only* (r, w) — so any *scalar* leg gives $c_u/c_d \equiv 1$. A scalar sees radius, not orientation. The u/d separation lives in the smallest *non-scalar* invariant of the $SU(3)_F$ composition: the exterior square $\Lambda^2 F$, i.e. the oriented anchor-plane area $\mathrm{Pl}_{\mathbf{1},a}(K)$. The "11" is then *generated*, not read:

$$(K, \mathbf{1}, a) \to B_K = \left(\begin{smallmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{smallmatrix}\right) \to \mathrm{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \to \|\mathrm{Pl}(K)\|_1 = 11,$$

$$\boxed{\frac{c_u}{c_d} = \frac{g_{\mathrm{car}}\,\|\mathrm{Pl}_{\mathbf{1},a}(K)\|_1}{N_{\mathrm{fam}}^2\,\Delta_Q} = \frac{5\cdot 11}{9\cdot 13} = \frac{55}{117}} \quad \left(\text{anchor microcode } = \tfrac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}\right).$$

Typing: [E] for the Plücker / microcode identities ($\mathrm{Pl}(K) = (-1, 6, 4)$, $\|\cdot\|_1 = 11$, $5\cdot 11/(9\cdot 13) = 55/117$); and — by Readout Rigidity (v49, below) — [E] also for the *physical* ratio $\mathcal{C}_{ud}(\rho^\star) =$ this $\Lambda^2 F$ readout, which is *constant* on the derived selector stratum and so does *not* need any point-selection on $\Lambda^2 F$. The quark residual is thereby re-typed from "find the right scalar y" to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale (U_{point}) stays as an anchor, no new scalar, no fishing.

Audit (the 121 *lemma, [*`verification/v119_review_validation_2.py`*]):* the "11" has a second, independent appearance [E]: with $h = \mathbf{1} + a = (2, 2, 3)$ in the canonical anchor ordering, $h^T R h = 121 = 11^2 = \|\mathrm{Pl}(K)\|_1^2 = (p_3 + 1)^2$ — the quadratic *anchor norm* of the residue operator R reproduces the Plücker integer (sensitivity control: the three orderings of h give $\{105, 121, 135\}$; only the canonical one yields 121). Bonus checksums: $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$ (the entry sum of R) and $a^T R a = 40 = p_1 p_3 = 10b_1 - 1$. Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. [`verification/v42_exterior_leg.py`]

Bridge test ([`verification/v43_exterior_bridge.py`**]).** The typing is confirmed: for the $SU(3)$ holonomy the exterior square is the conjugate representation, $\Lambda^2(\widetilde{M}) = \widetilde{\widetilde{M}}$ exactly — the exterior leg *is* the $\bar{\mathbf{3}}$. But the *continuous* action $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$ does *not* equal the *integer* $\mathrm{Pl}(K) = (-1, 6, 4)$: the integer Plücker is the combinatorial K datum. So the bridge $\rho^\star \to \mathrm{Pl}(K)$ is the *discrete* non-abelian-Hodge invariant (the integer R; $\det R = 8$ and $\mathrm{Spec}(Q_+) = \{1, 2, 3\}$ already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the $\Lambda^2 F$ side.

Lie-level grounding — no special math ([`verification/v44_carrier_exterior.py`**]).** The discrete invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier, $16 = \Lambda^{\mathrm{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$, and under $SU(5) \subset SO(10)$ the up quark sits in $10 = \Lambda^2(5)$ while the down sits in $\bar{5} = \Lambda^4(5)$. So the *exterior degrees* are $\deg(u^c) = 2$, $\deg(d^c) = 4$, with $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$ (the hexagon) and $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$ (the sheet). The "Λ^2" of the exterior leg is therefore not exotic Hodge data — it is the carrier's own Λ^2 grading (the up quark *lives* in $\Lambda^2(5)$). This re-types the residual once more: from "discrete non-abelian-Hodge invariant" to "the carrier exterior degree" — a standard [E] fact. *Honest scope:* this grounds the *type*; the Plücker *number* 55/117 stays the family $\mathrm{Pl}(K)$ [E], and the carrier-$\Lambda^2 \leftrightarrow$ family-$\Lambda^2 F$ link (via family $\otimes$ carrier $= 15 = 16-1$) plus the normalisation remain [C].

The "11" is Pascal — a third, simplest origin ([`verification/v45_family_exterior.py`**]).** Following that link, $\mathrm{Pl}(K)$ is the *exterior algebra of the family fundamental* $4 = \mu_4$ (the $A_3 = SU(4)$ fundamental): $\Lambda^k(4) = \binom{4}{k}$ is Pascal row $4 = (1, 4, 6, 4, 1)$ with full sum $2^4 = 16 = \dim S^+$, and $|\mathrm{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$ are exactly its distinct exterior powers. Hence

$$\boxed{\|\mathrm{Pl}(K)\|_1 = \textstyle\sum_{k=0}^{2}\binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\mathrm{car}}}$$

(the cumulative $\Lambda^\bullet(\mu_4)$ up to degree $2 = \deg(u^c)$), and family $\otimes$ carrier $= 15 = \dim \mathfrak{su}(4) = \Lambda^2(5) \oplus \Lambda^4(5) = 10+5 = N_{\mathrm{fam}} g_{\mathrm{car}}$ read three ways. So the "11" is generated a *third* way — the cumulative family exterior algebra up to the up-quark's degree — pure Pascal/branching, *no special math*. The [C] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

Lemma 6 (U5 — the "11" worry, *discharged* by Readout Rigidity)**.** *The original worry was that* $c_u/c_d = \mathcal{C}_{ud}(\rho^\star)$ *must be evaluated at the uniquely selected* $\rho^\star$ without *using* 11 *as input, so that until then* 55/117 *would be a mere table-level rational shadow.* ***This is now resolved (v49):*** $\mathcal{C}_{ud}$ *is* constant *on the discrete selector stratum* $\mathcal{S}_{8,Q}$*, so the "11" is* earned *as the rigid Plücker* *readout* $\|\mathrm{Pl}(K)\|_1$ *(v42)* — independently *of point-uniqueness. Point-uniqueness of* $\rho^\star$ *enters only the* absolute *amplitude scale, not the ratio.*

Lemma 7 (U6 — the four certificates)**.** $C_U^{(1)}$ *wall enumeration (done);* $C_U^{(2)}$ D_4*-fixed character solver;* $C_U^{(3)}$ *selector uniqueness (one* S*-equivalence class);* $C_U^{(4)}$ *amplitude extraction — the* ratios $R, Q, c_u/c_d$ *are read off the (derived) selector stratum (closed, v49/v71); only the* absolute $U_f^\star, c_q$ *normalisation is the anchor.*

Progress: an explicit valid flat bundle is realised (RH solve) ([**E**]). A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue A_0 (eigenvalues $\{0, \frac{1}{3}, \frac{2}{3}\}$, diag $= (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$) for which the Fuchsian system $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)}/(z - i^k)$ simultaneously satisfies: the cusp class at each puncture; the splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (exponents $\sum_k A_k = \mathrm{diag}(2, 1, 1)$); the *path-ordered* monodromy at infinity is trivial, $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$, i.e. $M_1M_2M_3M_4 = \mathbf{1}$; and the representation is *irreducible* — **case A** (non-trivial $SU(3)_F$ mixing, not the diagonal case B). So a valid (U_{wall}) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope:* this is one realising point (existence + case A), likely one of a residual family; the unique $\nabla_F^\star$ still needs the $\det R = 8$ selector and c_u/c_d still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary (H2). [`verification/v33_explicit_flat_bundle.py`]

H2 bridge probed ([`verification/v34_h2_bridge_attempt.py`]). The explicit per-puncture monodromies M_k (cusp class, $\prod M_k = \mathbf{1}$ to 10^{-5}) were computed; but the natural resolvent-dressed diagonal extraction ($|\mathrm{diag}\, M_k| = (0, \frac{1}{2}, \frac{1}{2})$ times the C_6 resolvent factor) does *not* reproduce the lepton amplitudes $(0.475, 1.107, 0.917)$. So the precise $\Gamma_{ij}^{\min}$ geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family $\rho^\star$) was the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for $55/117$. *Resolved since (v118): the correct extraction is the resolvent determinant of the* $W(A_3)$ *monodromy, not the diagonal dressing — the probe's failure is thereby explained, and the dictionary is exact.*

Theorem U — the four-way split (sharper than one monster gate) [E]/[O]

D_4 fixes the admissible chamber, not the physical point (the D_4-fixed locus is positive-dimensional, U3); the discrete selectors $\det R = 8$, $\mathrm{SNF}(R) = (1, 1, 8)$, $\mathrm{Spec}(Q_+) = \{1, 2, 3\}$ do the cutting. The gate splits into four pieces, three of them essentially closed:

- U_{unitary} [**E**]: parabolic degree 0 + polystable $\Rightarrow$ unitary (Mehta–Seshadri) $\Rightarrow$ $\Phi = 0$; the harmonic metric is the *finite* invariant form H ($M_k^\dagger H M_k = H$), not a Hitchin PDE ([`verification/v40_harmonic_metric.py`]).
- U_{H2} [**E**]: R, Q, K are read off the bundle (splitting type $= a$, $\det R = 8$, $\mathrm{Spec}(Q_+)$) ([`verification/v39_uwall_selectors.py`]).
- U_{Λ^2} [**E**]: **Readout Rigidity** — on the discrete stratum $\mathcal{S}_{8,Q} = \{\det R = 8, \mathrm{SNF}(R) = (1, 1, 8), \mathrm{Spec}(Q_+) = \{1, 2, 3\}\}$ the anchor-plane readout is *constant*, $\mathcal{C}_{ud}(\rho) = g_{\mathrm{car}} \|\mathrm{Pl}_{\mathbf{1},a}(K)\|_1/(N_{\mathrm{fam}}^2 \Delta_Q) = \frac{55}{117}$, *independent* of the continuous D_4 position — so c_u/c_d needs no point uniqueness ([`verification/v49_readout_rigidity.py`], [`verification/v42_exterior_leg.py`]).
- U_{point} [**O**]: full uniqueness of $\rho^\star$ — needed only for the *complete* $U_f^\star$ amplitude matrix, *not* for c_u/c_d.

Headline: the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; only U_{point} stays open.

The simple closure ([`verification/v71_simple_r_bridge.py`]). The selector stratum $\mathcal{S}_{8,Q}$ is now *fully derived*: $\det R = 8 = \mathrm{rank}\, E_8$ with $\mathrm{SNF}(R) = (1, 1, 8)$ is the lattice invariant (cf. $\det Q = 3 = N_{\mathrm{fam}}$, v70), and $\mathrm{Spec}(Q_+) = \{1, 2, 3\}$ is the D_4-equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants $(\det Q, \det K, \det R, \det L) = (3, 4, 8, 20)$, product $1920 = |W(D_5)|$. So by Readout Rigidity the quark *ratios* ($c_u/c_d = g_{\mathrm{car}} \cdot 11/(N_{\mathrm{fam}}^2 \cdot 13) = \frac{55}{117}$, etc.) are *pure integer Plücker readouts — no transcendental monodromy solve.* The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece, U_{point}, is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensionful scale, not a missing computation.

Gate 1 complete: $U_{\mathrm{point}} \to v_{\mathrm{geo}}$ **(**[`verification/v75_upoint_to_vgeo.py`]**).** The absolute amplitudes are not free either. The lepton amplitudes are exact ($c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, v20), every cross-sector c-ratio is closed (Plücker), and each *sector product* is a clean $\varphi_0^{\rm ret}$-power (Grand Mass Volume, v46: $\det M_{\rm sector} \sim (\varphi_0^{\rm ret})^{6,9,10}$; the lepton product is $\frac{16}{7}\cdot\frac{4}{3}\cdot\frac{7}{6} = \frac{32}{9} = 2^{g_{\rm car}}/N_{\rm fam}^2$). Since (pairwise ratios) + (product) ⇔ (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale* $v_{\rm geo}$. **So $U_{\rm point}$ is not an independent gate: it reduces to $v_{\rm geo}$, the *same* one irreducible dimensionful anchor as gravity's $1/G$ (v68) — the two [O] anchors collapse to one.**

Research Contract 2 — ($G_{\rm metric}$)

Goal. Construct the TFPT metric measure

$$Z[J] = \lim_{\chi\to\infty} \int_{\mathcal{G}_\chi/\mathrm{Diff}} \exp\big(-S_{\mathrm{rel},\chi}[g, A, \psi] + J\cdot O\big)\, d\mu_{\mathrm{Cal},\chi},$$

$$S_{\mathrm{rel},\chi} = \mathrm{Tr}\, f\Big(\tfrac{D_{\rm rel}+A_\Sigma}{\chi}\Big) - \mathrm{Tr}\, f\Big(\tfrac{D_{\rm ref}}{\chi}\Big) + \tfrac{i\pi}{2}\Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed $R + R^2$ equation.

Lemma 8 (G0–G1 — configuration space & BRST/Hodge Fredholm)**.** *For $s > 5/2$ a Sobolev space of Θ-symmetric metrics on the doubled seam collar M_Σ admits a gauge slice; the gauge-fixed gravitational complex $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2T^*M) \xrightarrow{K^*} \Omega^0(TM)$ with $\Delta_{\rm gh} = K^*K$ and $\Pi_{\rm phys} = \Pi_{\mathrm{BRST/Hodge}}\Pi_{TT}$ is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (*not *rhetorically).* Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).

Lemma 9 (G2 — finite relative spectral action, *heat-kernel grounded* [E])**.** *$S_{\mathrm{rel},\chi}$ is finite, differentiable, with a local Seeley–DeWitt expansion* $\mathrm{Tr}\, f(D^2/\chi^2) \sim f_4\chi^4 a_0 + f_2\chi^2 a_2 + f_0 a_4 + \cdots$*. For the Lichnerowicz Dirac operator $D^2 = \nabla^2 + \frac{R}{4}$ (so $E = -\frac{R}{4}$, spinor trace $\mathrm{tr}\,\mathbf{1} = 4$) the Gilkey coefficients give, per $(4\pi)^{-2}$,*

$$a_2 = \tfrac{1}{6}\,\mathrm{tr}(6E + R\,\mathbf{1}) = -\tfrac{R}{3} \quad (\textit{Einstein–Hilbert } \textstyle\int R), \qquad a_4\big|_{R^2} = \tfrac{1}{360}\,\mathrm{tr}(180E^2 + 60RE + 5R^2\mathbf{1}) = \tfrac{R^2}{72}$$

(the R^2/Starobinsky term), so the spectral action structurally *produces $S_{\rm eff} = \int\sqrt{g}\,\frac{\bar M_{\rm Pl}^2}{2}(R + \frac{R^2}{6M^2}) + \cdots$ with the scalaron mass ratio fixed by the cutoff moment, $M^2/\bar M_{\rm Pl}^2 = 6(4\pi)^2/f_0$. The TFPT closure $M^2/\bar M_{\rm Pl}^2 = c_3^7$ fixes $f_0 = 6(4\pi)^2/c_3^7$, i.e. $M = c_3^{7/2}\bar M_{\rm Pl} = 3.06\times 10^{13}$ GeV — the boundary normalisation c_3* is *the gravitational scale — and hence the Starobinsky slow-roll forms $n_s \simeq 1 - \frac{2}{N}$, $r \simeq \frac{12}{N^2}$. The $R + R^2$* structure *is convention-independent (standard Gilkey); the precise rational $\frac{1}{72}$ and factor 6 are the standard Dirac conventions used here ([verification/v36_spectral_action_g2.py], [verification/v28_gravity_fR.py]).*

The simpler resolution: "full QG" for physics is the gap-decoupled admissible sector ([E]). The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5): $2\|V_{\rm metric}\| = 0.785 < \Delta = 6\log\frac{3}{2} = 2.433$, so the admissible sector keeps a *strictly positive* effective gap $\Delta_{\rm eff} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

$$\boxed{\begin{array}{c}\text{admissible IR sector} = \text{closed under stated RP;}\\ \text{ambient metric measure (G6)} = G_{\rm metric}\ \text{gate}\end{array}}$$

G2 supplies the local action ($R+R^2$, heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce ($G_{\rm metric}$) to the ambient projective limit (G6). **Precise status (Alessandro's**

boundary): the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure remains a *strict TOE completion target.* Its absence does *not* affect the bounded IR claim, but it *does block certification as a strict physical TOE* — so we do not call it "mere mathematical completeness". [`verification/v36_spectral_action_g2.py`]

The glue Q-system: the index-4 object of the G_{net} statement exists explicitly ([E]/[C]). The G_{net} target — *"the seam–Calderón inclusion has Jones index $4 = |\mu_4|$"* — no longer refers to an unknown object. On the glue group algebra $A = \mathbb{C}[\mathbb{Z}_4]$ (multiplication $\delta_a \otimes \delta_b \mapsto \delta_{a+b}$, unit δ_0) *all* Longo Q-system axioms hold exactly [E]: associativity, unit law, the Frobenius identity $(m\otimes 1)(1\otimes m^*) = m^*m = (1\otimes m)(m^*\otimes 1)$, and specialness $mm^* = |\mathbb{Z}_4|\,\mathbf{1}$ — a special C* Frobenius algebra with

$$\boxed{\dim\theta = 4 = |\mu_4| = [B:A]}$$

exactly the Carrier Index Lemma value (KLM: $\mu_{\text{carrier}} = 16 = 4^2\mu_{E_8}$ re-verified). *Locality = isotropy [E]:* on the Lagrangian glue $\langle(1,1)\rangle$ the discriminant form gives $q(k(1,1)) = k^2 \in \mathbb{Z}$ for every k — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step [E]:* $\{0,2\} < \mathbb{Z}_4$ closes and gives the unique intermediate Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ tower step, with index factorisation $4 = 2\times 2$. *R2 sharpened [C] (recorded, not claimed):* what remains is the operator-algebraic *identification* "the seam–Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system" — matching two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification.

The hull is the graded module of the Q-system ([`verification/v128_graded_hull.py`]). The Q-system is not merely an abstract companion of E_8 — E_8 *realises* it [E]: the 240 roots, built as explicit norm-2 vectors in $D_5 \oplus A_3$+glue coordinates, decompose as $240 = 52+64+60+64$ over the glue $\mathbb{Z}_4$ (the v89 branching as vectors), and for *all* 6720 root pairs whose sum is again a root, $\text{coset}(r_1{+}r_2) = \text{coset}(r_1) + \text{coset}(r_2) \bmod 4$:

$$\boxed{E_8 \text{ is a } \mathbb{Z}_4\text{-graded Lie algebra over the carrier; the grading group is the glue = the Q-system}}$$

— the checksum hull is the Q-system's graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that E_8 already *carries.* [`verification/v125_glue_qsystem.py`]

Graded Frobenius ([`verification/v143_graded_frobenius.py`]). The identification holds at the finite Lie level: the graded module is in fact a graded *Frobenius* structure, i.e. everything the Q-system demands of its module is already exact inside E_8 [E]: (i) *coset duality* — $\text{coset}(-r) = -\text{coset}(r) \bmod 4$ for all 240 roots ($-C_1 = C_3$ bijectively, C_0, C_2 self-dual), so the Killing form restricts to a *nondegenerate* invariant pairing $g_k\times g_{-k} \to \mathbb{C}$; (ii) the glue character average $\frac{1}{4}\sum_k i^{k\,\deg}$ is *exactly* the carrier projector (the conditional-expectation shadow, quasi-basis index $|\mathbb{Z}_4| = 4$); (iii) each glue sector is *one* $W(D_5)\times W(A_3)$ orbit with multiplicity-one dimensions $64 = 16\cdot 4 = (\mathbf{16},\bar{\mathbf{4}})$, $60 = 10\cdot 6 = (\mathbf{10},\mathbf{6})$ — simple currents, sector ring $= \mathbb{Z}_4$ = the $\mathbb{C}[\mathbb{Z}_4]$ fusion, halfway subgroup $\{0,2\} = SO(16)_1$. *R2 residue [C] (final form):* exactly *one* conformal-net statement remains — the seam–Calderón inclusion of the 16-Majorana *net* carries this Q-system — with no remaining finite freedom.

Where that statement lives: the NS/R sector census ([`verification/v148_fock_census.py`]). The remaining work is scoped exactly. Every *untwisted* (Neveu–Schwarz) state of the 16 Majoranas has *integer* $(D_5, SO(6))$ weights, and integer weights lie in the even discriminant classes — so the NS module supports only $\{(0,0),(2,0),(0,2),(2,2)\}$: **the odd glue classes $k(1,1)$ — the actual μ_4 simple currents — have zero NS support; they are twisted (Ramond-type) modules.** The Ramond zero-mode module ($2^8 = 256$ ground states) carries exactly the four odd class pairs, 64 each, and splits $128+128$ into the odd sectors of the *two* Lagrangian glues $\langle(1,1)\rangle$ and $\langle(1,3)\rangle$ (v92): *choosing the glue is choosing the Ramond projection* (128 = one $SO(16)$ spinor chirality), and E_8 assembles as $248 = 120_{\text{NS}} + 128_{\text{R}}$. *Consequence [C]:* the index-4 extension cannot even be *stated* on the untwisted module alone; R2's one remaining statement is irreducibly about the twisted-sector structure of the seam–Calderón net, with the sheet choice realised as the R-projection; the finite shadow is exhausted (v113/v125/v143 + this census).

Lemma 10 (G3–G4 — reflection positivity, fixed and integrated)**.** *For each* Θ*-invariant* g *in the Calderón ball the physical boundary kernel* $\mathcal{K}_g = P_{\mathrm{adm}}\Pi_{\mathrm{phys}}\mathcal{N}_g^{TT}\Pi_{\mathrm{phys}}P_{\mathrm{adm}}$ *is reflection positive,* $\langle\Theta F, F\rangle_{\mathcal{K}_g} \geq 0$ *(G3); and* $d\mu_{\mathrm{Cal},\chi}$ *is* Θ*-invariant, supported where G3 holds uniformly, so* $\int\langle\Theta F, F\rangle_{\mathcal{K}_g}\, d\mu_{\mathrm{Cal},\chi} \geq 0$ *(G4).* ***G4 is the decisive step:*** *RP after integrating over dynamical metrics, not merely at fixed background.*

Lemma 11 (G5 — gap dominance [E])**.** $2\|V_{\mathrm{metric}}\|_{\mathrm{rel}} < \Delta = 6\log\frac{3}{2}$*, with* $\|V_{\mathrm{metric}}\|_{\mathrm{rel}} \leq 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$*, hence* $2\|V\| = 0.7852 < 2.4328$ *and* $\Delta_{\mathrm{eff}} \geq \Delta - 2\|V\| > 1.647 > 0$*. The numerical inequality is trivial (*[`verification/v29_research_contract_certs.py`]*); the work is the operator-norm bound (hard).*

Sugawara reading + atomic OS moment ([`verification/v171_os_moment_cluster.py`]**).** The G5 budget is the E_8 Sugawara constant: $248c_3^2 = \frac{31}{8\pi^2}$ with $31 = 1 + h^\vee(E_8)$ and $c((E_8)_1) = 248/31 = 8$, so gap dominance reads $|R^+(A_3)|\log(N_{\mathrm{fam}}/|\mathbb{Z}_2|) > \frac{1+h^\vee(E_8)}{4\pi^2}$ — the A_3 transfer gap beats the E_8 Sugawara metric budget. Independently, the G3 reflection positivity is a *positive moment problem*: $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ makes every two-point correlator $C(n) = A + B(64/729)^n + C(1/729)^n$ the moment sequence of a positive atomic measure, so each OS Hankel matrix factorises as a Vandermonde Gram and is positive semidefinite (a clean RP certificate; cluster constant 729/665, correlation length $\xi = 1/(6\log\frac{3}{2}) = 0.4111$).

Theorem 1 (Decoupling theorem — physical low-energy claims do *not* need the full G_{metric})**.** If *the admissible sector is reflection-positive (G4) and the metric coupling is gap-dominated,* $2\|V_{\mathrm{metric}}\| < \Delta = 6\log\frac{3}{2}$ *(G5),* then *the admissible IR measure is stable under the ambient completion: a strictly positive effective gap* $\Delta_{\mathrm{eff}} = \Delta - 2\|V\| > 0$ *protects the admissible sector from the (un-built) ambient/deep-UV, so every* physical low-energy *read-out (masses, mixings,* α^{-1}*, the* $R{+}R^2$ *regime) is independent of the open ambient measure. Symbolically*

$$\boxed{\mathrm{RP}\ +\ 2\|V\| < \Delta \implies \textit{admissible IR measure stable under ambient completion}\ .}$$

Thus (G_{metric}) *is* not *a diffuse "full quantum gravity missing" block: it is the sharply delimited* ambient projective limit *(G6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on.* ***Typing:*** *the inequality is machine-checked [E]; the operator-norm bound on* $\|V_{\mathrm{metric}}\|$ *is the conditional input.*

Lemma 12 (G6–G7 — projective limit & Ward identities)**.** *A projective system* μ_{χ_n} *with* $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$*, uniform moment bounds* $\sup_n\int\|g\|_{H^s}^p\, d\mu_{\chi_n} < \infty$*, tightness and regulator-independence yields* $\mu_{\mathrm{QG}} = \lim\mu_\chi$ *on* $\mathcal{G}/\mathrm{Diff}$ *(G6, the hardest step); the limit obeys the BRST Ward identities* $\langle s\mathcal{O}\rangle = 0$*, whence* $\nabla^\mu T_{\mu\nu} = 0$ *and the* η *boundary phase cancels the APS anomalies (G7).*

Gate 2 reduction: G6 is a *boundary* projective limit, not a bulk one ([`verification/v76_gmetric_reduction.py`]**)**

The exact decoupling margin is $\Delta_{\mathrm{eff}} = \Delta - 2\|V\| = 6\log\frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$, with $\|V_{\mathrm{metric}}\|_{\mathrm{rel}} \leq 248\, c_3^2 = \dim(E_8)\, c_3^2 = \frac{31}{8\pi^2}$ $(31 = 2^{g_{\mathrm{car}}} - 1)$. **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so G6 is a *boundary* projective limit. The conditional theorem is then

$$\boxed{\mathrm{RP(seam\text{-}Calder\acute{o}n\ boundary\ kernel)} + \mathrm{tightness} \implies G_{\mathrm{metric}}\ \mathrm{closes}}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure. **Residual:** the constructive boundary projective limit (constructive-QFT grade) — reduced, not closed. So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) reduced to a boundary measure, no longer a diffuse "full QG missing". [C]/[O]

And that boundary measure is a *rigorously constructed* net ([verification/v77_e8_conformal_net.py]). The level-1 central charges $c = \dim \mathfrak{g}/(1+h^\vee)$ give $c(E_8)_1 = \frac{248}{31} = 8 = \operatorname{rank} E_8$, $c(D_5)_1 = \frac{45}{9} = 5$, $c(A_3)_1 = \frac{15}{5} = 3$, and $c(D_5)+c(A_3) = 5+3 = 8 = c(E_8)$ is a *conformal embedding* $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$ (coset $c = 0$). Now $(E_8)_1$ is the holomorphic $c = 8$ E_8*-lattice* VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete:** *identify the seam-Calderón boundary measure with the* $(E_8)_1$ *lattice net* (carrier $= (D_5)_1 \times (A_3)_1$ subnet), and the gap $\Delta_{\text{eff}} > 0$ supplies the clustering/tightness — so $G6$ is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [E] (charges, embedding) + [C] (identification)

The Simple-Current Extension Theorem — the central G_{net} closing statement [E]/[C] ([verification/v154_simple_current_theorem.py])

The conditional theorem can now be stated as one clean *simple-current extension.* Let $A = (D_5)_1 \otimes (A_3)_1$ be the carrier net and $L = \langle(1,1)\rangle \subset \mathrm{Disc}(D_5) \oplus \mathrm{Disc}(A_3) = \mathbb{Z}_4 \oplus \mathbb{Z}_4$ the isotropic μ_4 glue line ($q(k(1,1)) = k^2 \in \mathbb{Z}$). Then the local extension $B = A \rtimes L$ has

$$\boxed{[B : A] = |L| = 4 = |\mu_4|, \quad c(B) = 5{+}3 = 8, \quad \mu(B) = \frac{\mu(A)}{|L|^2} = \frac{16}{16} = 1}$$

so B is *holomorphic* (KLM μ-additivity), and a holomorphic c=8 chiral net is the lattice net of the *unique* even-unimodular rank-8 lattice — hence $\boxed{B \cong (E_8)_1}$ (the same-c rival $SO(16)_1$, μ=4, is excluded). Every algebraic ingredient is machine-checked: the $\mathbb{C}[\mathbb{Z}_4]$ Longo Q-system ([verification/v125_glue_qsystem.py]), the $\mathbb{Z}_4$-graded Frobenius E_8 hull ([verification/v143_graded_frobenius.py]), the Ramond glue sectors ([verification/v148_fock_census.py]). **Consequence:** G_{net} is internally *closed* if the seam–Calderón boundary net is *defined* to be A; it stays the *one* external identification "the RP seam–Calderón net is A" if the boundary net is to be *derived* from the seam kernel. Either way this is the single strict-TOE certification theorem still carrying weight — a clean extension statement, not "find a quantum gravity".

And that one identification reduces to a single shared premise ([verification/v155_quasifree_boundary.py]). The identification splits (v110) into (a) "the seam Calderón involution is sheet-odd" — the double-cover deck, the same $|\mathbb{Z}_2|$ that halves $c_3 = \frac{1}{2\cdot 4\pi}$ (structural) — and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not independent*: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the bulk covariance, and the compression of a contraction ($i\Gamma$ with spectrum in $[-1,1]$) is a contraction — so the boundary/seam state is automatically quasi-free, with Wick inherited (every boundary correlator a Pfaffian of Γ_d). Hence (b) *follows* from "the seam bulk is a reflection-positive free field" — the *same* premise already load-bearing in the area law ([verification/v59_area_law_evidence.py]) and the EH mechanism ([verification/v150_replica_eh_model.py]/[verification/v151_bfk_split.py]). **So the entire seam-side residual is one physical premise** — the seam bulk is a free RP field — shared by G_{net}, the area law and R3, and given it the chain Gaussian bulk $\Rightarrow$ quasi-free boundary $\Rightarrow$ one kernel = net (v113) $\Rightarrow$ carrier $A \Rightarrow (E_8)_1$ is exact. It is not reducible by a finite computation — it is the one place TFPT meets functional analysis.

The construction, exhibited and verified ([verification/v156_seam_net_construction.py]). The "$B \cong (E_8)_1$" step is no longer an assertion — the chiral net is built explicitly and the identification *checked.* The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian acts on a boundary mode e^{ikx} by the harmonic extension $e^{ikx-|k|y}$: $-\partial_n u|_0 = |k|\, u$, so for a free bulk the seam operator is the massless chiral dispersion $\omega_k = |k|$ that defines the free-fermion net on S^1. Sixteen Majoranas give $c = 8 = 5{+}3$; the weight-1 currents are $248 = 120_{\text{NS}} + 128_{\text{R}}$ (the $SO(16)$ bilinears plus the Ramond spinor, Frenkel–Kac); and the holomorphic character is

$$\boxed{\chi = \frac{E_4}{\eta^8} = q^{-1/3}\big(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + \cdots\big) = j^{1/3}}$$

(verified by q-series), the 248 at level 1 being $\dim E_8$ — so the net *is* $(E_8)_1$. *What remains* is exactly (i) the free-bulk premise above and (ii) the operator-algebraic existence / uniqueness of the net generated by the seam Calderón kernel (a Kawahigashi–Longo-type net-construction theorem). Route 3 thus *exhibits and verifies* the target; the irreducible residual is the one premise + the standard net-existence step — where TFPT meets constructive QFT, now sharply located.

And the free-bulk premise itself reframes: freeness is a rigid fixed point, not a postulate ([verification/v157_rigid_fixed_point.py]). Three facts click together. (i) The DtN/Calderón symbol is *universally* $|k|$: the boundary operator of *any* second-order elliptic bulk is an order-1 ΨDO with principal symbol $|\xi|_g$ (Lee–Uhlmann), so the free chiral dispersion is bulk-detail-independent and any interaction lives in the lower-order (irrelevant) symbol. (ii) A holomorphic c=8 theory is *rigid*: every field has weight $(h, \bar{h}{=}0)$, an exactly-marginal deformation needs $(1, 1)$, and with $\bar{h}$=0 *no* $(1, 1)$ field exists — the fixed point is isolated, there is no continuous knob along which a non-Gaussian interaction could turn on. (iii) The *same* $|\mathbb{Z}_2|$ that halves the one-sided S^2 Gauss–Bonnet ($c_3 = \frac{1}{|\mathbb{Z}_2|\,2\pi\,\chi(S^2)} = \frac{1}{8\pi}$) labels the sheet *and* is the Ramond/GSO projection giving μ=1 ($248{=}120_{\mathrm{NS}}{+}128_{\mathrm{R}}$) — one object, three roles, so the geometry fixing c_3 is the projection making the net holomorphic. **Consequence:** premise (A) shrinks from "prove the seam is Gaussian" to "the gapped ($\Delta{=}6\log\frac{3}{2}{>}0$) boundary flow reaches the holomorphic fixed point" — given which, freeness is *forced* by rigidity + uniqueness ($(E_8)_1$, v83), not fine-tuned. A strictly smaller and simpler residual.

And the destination is a *proven* stable fixed point ([verification/v158_fixed_point_stable.py]). An exact operator-dimension count settles it: the chiral Majorana has $h{=}\frac{1}{2}$, the bilinear current h=1, the quartic h=2. The relevant window $0 < h < 1$ for bosonic deformations is *empty* (bosonic chiral operators have integer h, lowest 1) — nothing relevant flows the free theory away; the h=1 currents are chiral (v157), not $(1, 1)$-marginal; and the lowest genuinely-interacting operator (the quartic, $h{=}2{>}1$) is *irrelevant*, as are all higher $2k$-fermi operators ($h{=}k$). So the free chiral c=8 fixed point is *isolated and stable* against all interactions. Premise (A) thereby reduces to a *basin-of-attraction* statement with a proven stable target — the gapped ($\Delta{=}6\log\frac{3}{2}$) flow lands here; the convergence is conformal-perturbation analysis, the destination is settled.

And the closure can be stated as one fixed-point theorem — which relocates (A) to a single internal premise ([verification/v160_seam_gaussianity_from_pf.py]). Phrase the argument internally: (i) a quasi-free state has *all* higher connected cumulants κ_{2n}=0 (the signed set-partition/Pfaffian inversion of $\langle\psi_1\cdots\psi_{2n}\rangle = \mathrm{Pf}(\Gamma)$ leaves only the degree-2 kernel — the v113 "one kernel = the net" property at the cumulant level); (ii) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (iii) a gapped primitive Perron–Frobenius transport has a *unique* fixed ray, so a non-Gaussian piece is impossible — if κ_{2n} were fixed it would be a *second* fixed ray (killing uniqueness), and if not fixed it sits in the gapped complement and decays to 0 (any gap > 0 suffices; the value $(2/3)^6$ only sets the rate). *The honest residual:* v56's gapped PF uniqueness is established on the *three-dimensional* scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (the flavour gap / horizon Page-recovery rates), whereas the 16-Majorana seam state already carries $\binom{16}{4}$=1820 and $\binom{16}{6}$=8008 higher-cumulant directions — the Schwinger cone the transport must contract has dimension $\gg 3$, and uniqueness must hold there (not merely on the quasi-free sub-cone) for the argument to be non-circular. **So (A) is equivalent to one strictly smaller, internal, exact-testable statement:** the admissible TFPT seam transport is a *primitive, spectrally gapped, RP-cone-preserving* map on the *full* Schwinger cone, not only on the scalar readout spectrum of v56 — which the suite does *not* yet establish [E]/[O].

And that full-cone premise is now *discharged* to a single finite one-particle statement ([verification/v161_one_particle_reduction.py]). The lever is Bogoliubov second quantisation: a quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dimensional Majorana space H_Σ, and the Fock space grades by correlator degree, so $\Gamma(t)$ acts on the degree-m sector as the exterior power $\Lambda^m(t)$. Three exact consequences follow (all machine-checked): (i) $\mathrm{spec}(\Gamma(t)|_{\deg m})$ is the set of products of m one-particle eigenvalues — the whole correlation spectrum is generated by the finite one-particle spectrum; (ii) **the cone gap *equals* the one-particle gap**: the sub-leading eigenvalue of $\Gamma(t)$ over *every* degree is $(2/3)^6$ (pad the largest contracted mode with fixed factors), so premise 5 on the cumulant space *is* premise 5 on the 16-dim space, the same number; (iii) the eigenvalue-1 sector is Λ^* of the fixed subspace — the disconnected Wick powers of K_Σ — so with a strict one-particle gap no independent connected cumulant is fixed, and an independent fixed cumulant *on* the fixed subspace is excluded by v113 (one polarisation ⇔ one vacuum). Premise 1 (quasi-freeness) is itself *forced*, not assumed: the $120 = \dim\mathfrak{so}(16)$ weight-1 chiral currents *are* the Bogoliubov generators on the 16 Majoranas, and v158 makes every non-Gaussian deformation irrelevant. **So the three full-cone premises collapse to one finite, 16-dimensional statement:** the seam RG transport's one-particle symbol is the gapped Calderón Bogoliubov map — fixed eigenspace = the rank-8 K_Σ polarisation (v113), sub-leading gap = $(2/3)^6$ (v56). Every *number* in it is already supplied by the suite; the one irreducible step is the physical identification that the actual seam RG generator equals this Bogoliubov map — the same operator-algebraic (Kawahigashi–Longo-class) statement as the net-existence residual of (A_2), now with the infinite-dimensional cone eliminated [E]/[O].

And that finite statement carries no *new* open content: it factors into the two already-open items ([`verification/v162_seam_transport_identification.py`] **).** Take the one-particle symbol apart into its two data, the gap value (β) and the fixed space (α). (β) *is forced*: a μ_4-equivariant operator is diagonal in the cusp basis — the deck average is the diagonal projection, $\sum_{k=0}^{3} U^k X U^{-k} = |\mu_4|\,\mathrm{diag}(X)$ for $U = \mathrm{diag}(1, i, -i)$, and the commutant of U is exactly the diagonal algebra — so its spectrum is the cusp weights $\mathrm{spec}(A_0^\star) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (the parabolic residue, v115), and the transfer eigenvalue $(1-\alpha)^{p_2}$ with p_2=6=$|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — every entry carrier data, no free knob; the lift to the 16 Majoranas is the established A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of the ω_k=$|k|$ free dispersion (v113/v156), the IR conformal free-fermion vacuum. **What is left** is only that the seam RG generator *is* μ_4-equivariant and admissible — which is the R1 "seam clock" identification (the transfer operator as the seam's gapped generator) — together with the A_2 net-existence theorem. Both are pre-existing open items, so premise (A) introduces *no independent open content*: it factors exactly into $A_2 + R_1$ with every number forced. The "free-bulk cloud" is gone; what remains is the same two identifications the programme already names — (A) is closed *conditionally* on them (a unification, honestly not an unconditional proof) [E]/[O].

And both residual identifications are pinned to their floor, so (A) has no independent open gate left ([`verification/v165_premise_a_closure.py`] **).** A_2 (net existence) is not open research but *assembly* of established mathematics: the 16 free Majoranas define a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic c=8 lattice net *is* $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125); the one TFPT-specific input — that the seam Calderón kernel is the free-fermion two-point function — is done because the DtN symbol is universally $|k|$ (v156/v157). So A_2 re-types [O] → [C] (closed by assembly). R_1 (the seam clock) reduces to `GATE.QGEO`: a μ_4-equivariant flat gapped generator on the carrier is *unique* (= $A_0^\star$, v115/v116), so its gap $(2/3)^6$ is forced once the seam generator lies in that class, and the residual is exactly one geometric identification — the seam collar boundary is the μ_4-punctured sphere $\mathbb{P}^1 \setminus \mu_4$ — whose cohomological half is already established (v137, b_1=3=N_{fam}). That is `GATE.QGEO`, an already-open realisation premise, *not* a new gate. By the No-Unit Theorem (v153) the irreducibles stay exactly $\{\pi, v_{\mathrm{geo}}\}$ — closing (A) adds none. **Final accounting:** premise (A) = (A_2 assembly, [C]) + (R_1=`GATE.QGEO`, already open) + the $\{\pi, v_{\mathrm{geo}}\}$ irreducible core (a theorem) — it carries *zero* independent open gates and ceases to be its own gate [E]/[O].

And the two structural residuals — net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the one irreducible open premise ([`verification/v175_net_existence_full_cone.py`] **).** The CAR second-quantisation functor $\Gamma(t)$=$\bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the *complete* Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ (dim 2^{16}=65536); by the exterior-power spectrum theorem every eigenvalue of $\Gamma(t)$ is a subset product of spec(t), so checking all 2^{16} products shows $\Gamma(t)$ is a *contraction* (full-cone RP), with eigenvalue-1 multiplicity 2^8=256 (the wedges of the rank-8 K_Σ polarisation) and sub-leading eigenvalue exactly the one-particle gap $(2/3)^6$ — so full-cone RP/gap reduce *for every m* (not only $m \le 6$, v161) to the one-particle contraction (Shale–Stinespring), *discharging the v160 scope premise by a theorem.* And A_2 is now an *assembled and verified* certificate: the $(E_8)_1$ character E_4/η^8=1+$248q$+$4124q^2$+$34752q^3$+$213126q^4$+$1057504q^5$ (level-1=248), the embedding 248=120+128, and the E_8 Cartan matrix even with det 1 (the unique rank-8 even unimodular lattice, v83) — the net *exists* by cited theorems. **Both residuals then collapse to the *same* input**: that the actual seam Calderón kernel is this gapped one-particle contraction on $\mathbb{P}^1 \setminus \mu_4$, i.e. `QGEO.REALIZE.01`, whose finite half is proven (v168). That realisation half is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]— the single irreducible structural premise; net existence and full-cone RP are [E].

The No-Unit Theorem — v_{geo} is an irreducible metrology primitive [E]/[O] ([`verification/v153_no_unit_theorem.py`])

v_{geo} is not an open gap. Every load-bearing compiler datum — $g_{\mathrm{car}}, |\mu_4|, N_{\mathrm{fam}}, \mathrm{rank}\, E_8, \alpha^{-1}$, the determinant ladder (3, 4, 8, 20), $c_3 = \frac{1}{8\pi}$, φ_0^{ret} — is mass-dimension 0 and *invariant* under a unit rescaling $L \mapsto \lambda L$, whereas a mass / energy / $1/G$ carries nonzero dimension. A map from dimension-0 inputs to a dimension-$d \neq 0$ output is simultaneously invariant and covariant: impossible unless the unit is introduced. **So a dimensionless boundary compiler provably cannot select an absolute scale** — v_{geo} is a *forced input.* The three apparent residuals then collapse onto the one unit, differing only in mass-dimension:

$$U_{\mathrm{point}} \sim v_{\mathrm{geo}}, \qquad \tfrac{1}{G} \sim v_{\mathrm{geo}}^2, \qquad m/\mu = e^{3/4} \text{ (a pure ratio)},$$

so the irreducibles stay exactly {one dimensionful unit, π}. *Re-typing:* v_{geo} is an *irreducible metrology primitive* (a declared unit, like choosing a metre), not a missing derivation.

Theorem G — the closure statement [O] (target)

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure μ_{QG}. Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT $R + R^2$ seam action

$$f_R R_{\mu\nu} - \tfrac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu}\Box - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\mathrm{Pl}}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: open; the deepest item. G5 certified; the regime equation closed [`verification/v28_gravity_fR.py`].*

Certifiability and recommended order

Building block	Machine	Hardness
W_{wall} enumeration ($C_U^{(1)}$)	Lean / Python / Wolfram	easy (*done*)
D_4 fixed-locus polynomial ($C_U^{(2)}$)	Sage / Mathematica	medium
selector uniqueness ($C_U^{(3)}$)	interval arithmetic	medium
SNF, determinants, spectra	Lean	easy–medium
principal-symbol ellipticity (G1)	xAct / Cadabra / Lean	medium
gap inequality (G5)	Lean	easy (*done*)
operator-norm bound (G5)	analysis + numerics	hard
projective limit (G6)	not primarily machine	very hard
Ward identities (G7)	BV/BRST formal + checks	hard

Recommended order.

$$\boxed{\text{Selector-triangle pairings} \;\rightarrow\; v_{\mathrm{geo}} \text{ scale anchor} \;\rightarrow\; G_{\mathrm{net}} \text{ inclusion theorem}}$$

The flavor interface (historically U_{wall}) is now reduced to the *selector-triangle pairings* (the dual normal pair (d, n) forcing R columnwise, v136/v139); it is finite, falsifiable and the cheapest visible upgrade. The v_{geo} scale anchor is the one dimensionful input (same nature as $1/G$). G_{net} — the index-4 seam-net inclusion certifying the metric sector — is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit the genuine summit. F_{transfer} (source→pole / relic / cosmology) is the remaining downstream interface.